**People's Democratic Republic of Algeria**
**Ministry of Higher Education and Scientific Research**
**Higher School of Computer Science**
**8 Mai 1945 - Sidi Bel Abbes**

# Graduation Thesis

To obtain the diploma of **Engineering Degree**

Field of Study : **Computer Science**

Specialization : **Computer Systems Engineering (ISI)**

## Theme

# Design and Deployment of a Digital Twin for the Optimization of an Intelligent Irrigation System in Precision Agriculture

Presented by
**Lebaili Ouail and Senhadji Said Mohamed**

Defended on : **July, 2026**
*In front of the jury composed of*

| | |
|---|---|
| **Mr. [Jury Member Name]** | President of the Jury |
| **Mr. [Jury Member Name]** | Thesis Supervisor |
| **Mr. [Jury Member Name]** | Co-Supervisor |
| **Mr. [Jury Member Name]** | Examiner |

*Academic Year : 2025/2026*

# Abstract

***Keywords***— keyword 1, keyword 2

# Table des matières


**1 Introduction** **5**

## I Introduction 6

**2 Introduction** **7**

2.1 General Introduction . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . 7

# Table des figures

# Liste des tableaux

# Chapitre 1

# Introduction

# Première partie

# Introduction

# Chapitre 2

# Introduction

## 2.1 General Introduction

Cyber threats continue to evolve in scale and sophistication, impacting private organizations as well as public institutions. Large-scale incident analyses show that modern attacks remain effective because adversaries repeatedly exploit common initial access vectors and operational weaknesses, leading to significant business and mission impact [ **?**].

In this context, penetration testing is widely used to evaluate security by simulating real attack behavior under controlled conditions. According to NIST, penetration testing verifies the extent to which a system, device, or process can resist active attempts to compromise its security, and often involves combining multiple weaknesses to gain deeper access than any single vulnerability would allow [ **?**, **?**]. Methodological guidance such as NIST SP 800-115 further emphasizes structured planning, execution, analysis of findings, and mitigation-focused reporting [ **?**].

At the same time, recent advances in large language models (LLMs) and agentic AI have enabled systems capable of planning and executing multi-step technical tasks. Research has begun exploring how multi-agent LLM frameworks can emulate the collaborative workflow of penetration testing teams to improve efficiency and reduce manual effort [ **?**].

Therefore, this project aims to leverage a hybrid multi-agent approach to support and improve the penetration testing process, while enforcing strict control over scope, safety, and traceability of actions and results. The goal is to increase the repeatability and scalability of testing activities without compromising governance and operational constraints. [1]